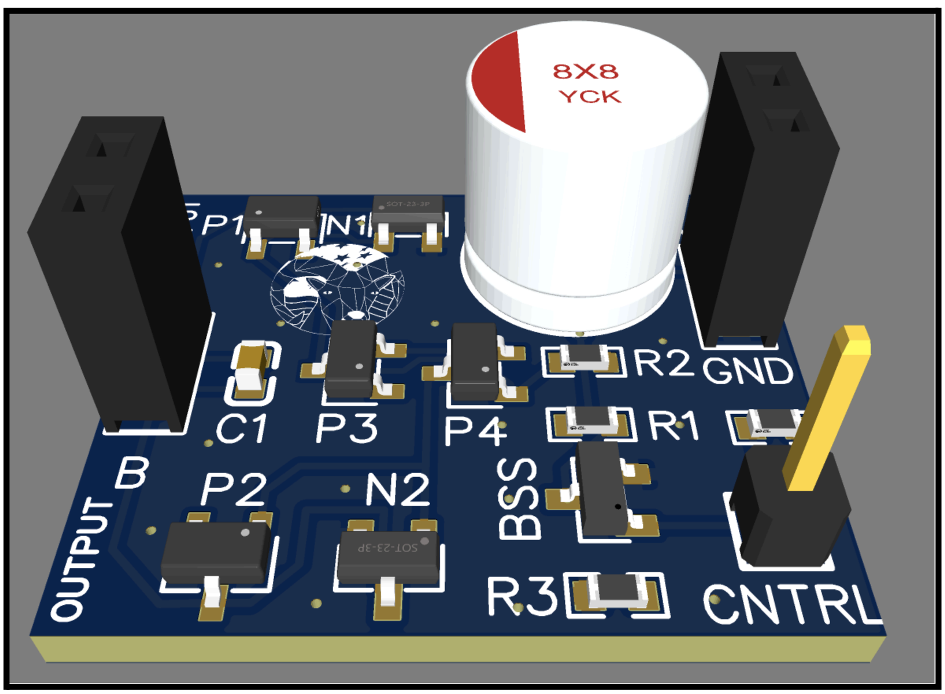
ARAM-DBH_0_13-0S Arduino/Esp32 Logic Level H-Bridge

This H-Bridge is a modern MOSFET-based design from ARAM, American Robotics Assisted Manufacturing. This circuit will allow you to control the output polarity of the H-Bridge using by toggling the CNTRL pin between logic Low and High. This H-Bridge can support up to a 12V input and can **output up to 2.1Amps continuously** or up to **2.5-3 Amps with the addition of a heatsink**. Amperages as high as **5 Amps surge current**. The ***inbuilt logic prevents shoot-through***, it is impossible to turn both sides of the H-Bridge on simultaneously. This MOSFET-based design keeps an **efficiency of 75-95%** within operating parameters. This is ideal for small microcontrollers such as **arduino or esp32**. The circuit also has a quick rise and **fall time of less than 35 nanoseconds.**



All Items Tested at 18C, with no heatsink or external airflow, unless otherwise specified

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Standby/Quiescent Current: < 0.001 Amps

Voltage Inputs

Voltage Input	Voltage
Max	12 Volts
Min	5 Volts

Logic Inputs

Logic Input	Supported
1.8V	*Yes * (Decreased Efficiency, Increased Rise/Fall Timing, 0-50C)*
3.3V	Yes
5V	Yes

Timing

Edge Timing	Time (nanoseconds)
Rise	< 35ns
Fall	< 35ns

Current Ratings

Current Ratings	Current (Amps)	Maximum Time (seconds)
Continuous	2.1 Amps	-
Continuous (With Heatsink)	2.5-3 Amps	-
Extended	2.5 Amps	180 Seconds
Burst	4 Amps	20 seconds
Peak	5 Amps	5 seconds

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Rds(On) and Efficiency:

Voltage (V)	Current (Amps)	RdsOn (Ohms)	Efficiency (%)
5	1.022	0.88	82
5	1.624	0.97	68.4
5	2.104	1.01	57.4
7	0.428	0.86	94.71
7	1.774	0.92	76.57
7	2.542	1.02	62.85
9	0.549	0.86	94.77
9	1.802	0.92	81.66
9	2.227	0.96	76.33
9	2.73	1.05	68
11	0.672	0.86	94.72
11	2.165	0.94	81.54
11	2.767	0.95	76.45
11	3.325	0.82	75.09
12	0.735	0.86	94.75
12	2.35	0.95	81.41
12	2.905	0.98	81.41
12	3.532	1.07	68.5

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Pinout Table

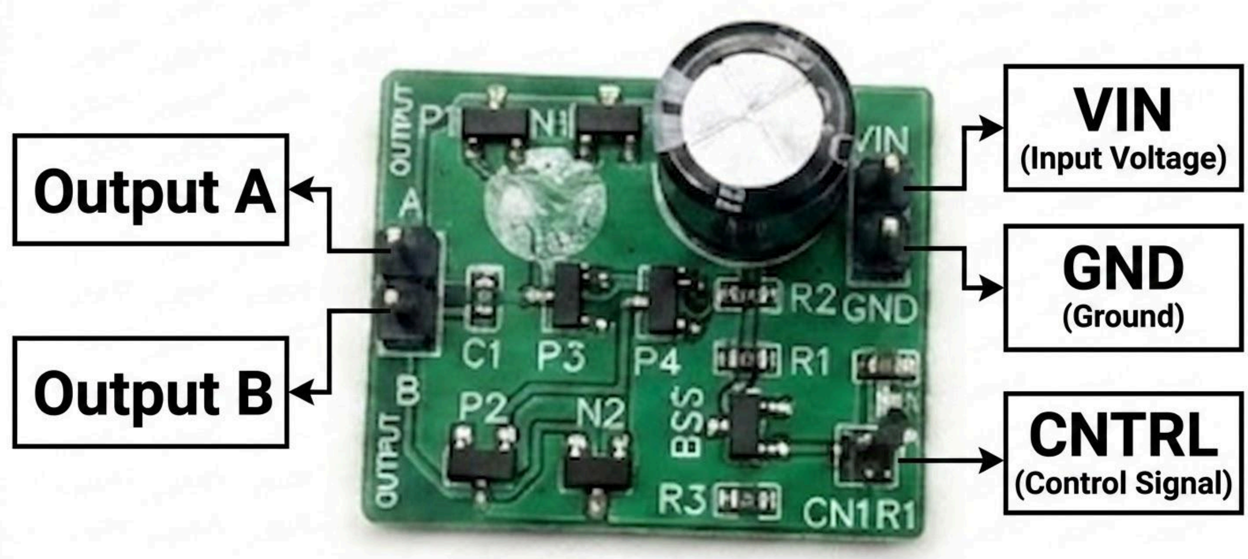
Pin Name	Use
Vin	Connect to Vin (5V-12V)
Gnd	Connect to GND (0V)
Cntrl	Toggle between logic HIGH (1.8V*/3.3V/5V) and LOW to set the polarity of the H-Bridge. If this pin is left floating the circuit will default the logic LOW polarity.
Out A	Output of H-Bridge, form circuit with Out B
Out B	Output of H-Bridge, form circuit with Out A

Physical Dimensions

28 mm x 40 mm

Weight

~4g



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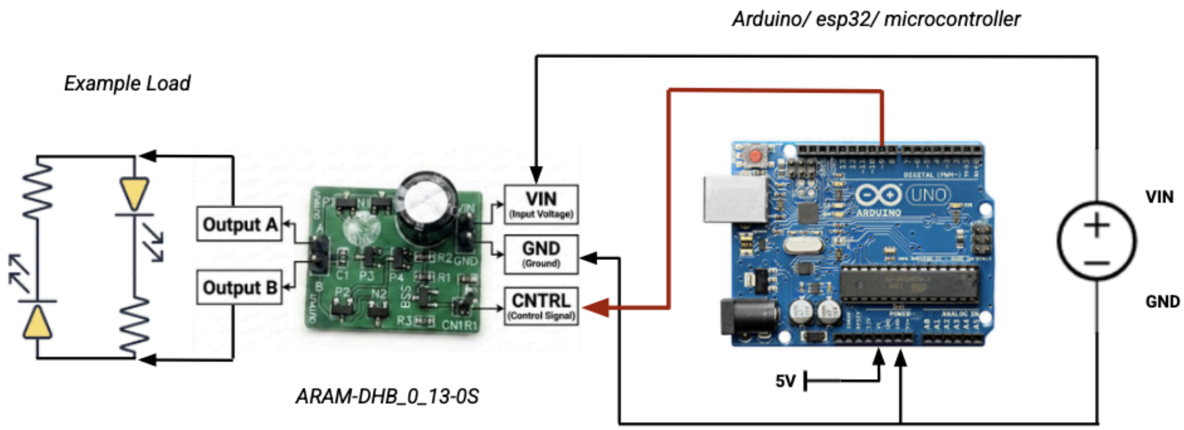
Example Usage



This example shows the polarity switching capabilities of the h-bridge when paired with a microcontroller. Vary switchFrequency to switch faster or slower.

Example Code:

```
//ARAM-DHB_0_13-0S      Example Code                rev: 4/25/2026

#define CNTRL_PIN 4      //change me to your desired cntrl pin
#define MICROS_TO_SECONDS (1000000)
uint32_t timeClock;
uint32_t nextSwitch;
short state = 0;
int switchFrequency = 10; //change me to your desired switching frequency (1/seconds)
double timeStep = MICROS_TO_SECONDS/switchFrequency;

void setup() {
 pinMode(CNTRL_PIN,OUTPUT);
 digitalWrite(CNTRL_PIN,HIGH);
 nextSwitch = micros();
}

void loop() {
 timeClock = micros();
 //every time we want to switch polarity simply shift the control pin from low to high or vise versa
 if(timeClock > nextSwitch){
   nextSwitch = timeClock + timeStep;
   state = !state;
   if(state) digitalWrite(CNTRL_PIN,HIGH);
   else digitalWrite(CNTRL_PIN,LOW);
 }
}
```

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Sincerely:

Jack Serlin,
Founder of American Robotics Assisted Manufacturing

Contact Email:

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